Robust Multiplicative VCG for Multi-Agent Failure Attribution:

Empirical Diagnosis, Mechanism Refinement, and the Aggregation–Selection Dichotomy

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Executive Summary

Main result. Within the multiplicative-VCG framework of the companion theory note, we identify two regime-specific failure modes of the original squared-loss mechanism on the Who&When benchmark, and we propose two refined mechanisms that significantly improve over the original VCG on both subsets. Table 1 reports the headline numbers.

Table 1: Step-attribution accuracy on Who&When. **Bold** marks the best practical (non-oracle) method per row. "Original VCG" uses $R = 0$ (no OMD calibration) and $\varepsilon = 10^{-6}$; see Section 4 for the relationship to Tong's earlier $R = 5, \varepsilon = 0.10$ numbers.

		Single discriminator (*oracle reference*)			Aggregation / mechanism (*practical*)			
Subset	n	Gemini	Qwen	GPT-OSS	Mean	Orig. VCG	L1-Huber	Conf-Select
HC	55	14.5%	10.9%	16.4%	12.7%	9.1%	**23.6**%	14.5%
AG	126	38.1%	27.0%	19.0%	33.3%	27.8%	30.2%	**35.7**%

What works on HC. The **L1-Huber VCG** (Section 3, kernel $\kappa_c(x) = 1 - \min(|x|, c)$, $c = 0.05$) achieves 23.6% on Hand-Crafted traces, a statistically significant improvement over the original VCG (+14.5pp, McNemar $p = 0.022$) and over Mean (+10.9pp, $p = 0.031$). The improvement is concentrated in the *maximal-disagreement* regime (60% of HC traces): on the 33 traces where the three discriminators' argmax-steps all differ, L1-Huber achieves 15.2% versus 9.1% for Median and 3.0% for Mean. The cross-step reputation-weighted median update of L1-Huber recovers signal that all per-step aggregators miss.

What works on AG. The **Confidence-Select VCG** (Section 3, $\hat{t}^* = \arg\max_t \hat{\theta}_t^{k^*}$ where $k^* = \arg\max_k \max_t \hat{\theta}_t^k$) achieves 35.7%, the highest accuracy among practical methods on Algorithm-Generated traces. It significantly improves over original VCG (+7.9pp, $p = 0.099$) and marginally exceeds Mean (prob) at $\varepsilon = 0.10$ (+0.8pp). The gap to the single-discriminator oracle ceiling (−2.4pp from Gemini's 38.1%) is not statistically significant ($p = 0.51$).

Why AG has a ceiling, and why two mechanisms instead of one. The two best mechanisms above are structurally different (aggregation vs. selection) rather than two settings of a single parametric family. We demonstrate this empirically in Section 5 via two unification attempts that both fail in informative ways:

- A *cross-trace VCG-payment prior* systematically nominates the wrong discriminator on AG — it picks D_1 (Qwen) because Qwen's reports are closest to the consensus on most traces, whereas D_0 (Gemini) is the most *accurate* discriminator. The resulting π-select achieves only 27.0%, exactly Qwen's solo accuracy. Any payment-based prior defined relative to consensus is biased toward consensus-followers and away from minority correct experts.
- A *within-trace evidence prior* $w_k^t = (\max_t \hat{\theta}_t^k)^\alpha \cdot \prod_{t' \neq t} \kappa_c(\cdot)$ does not interpolate smoothly between L1-Huber and Confidence-Select. Scanning $\alpha \in \{0, 0.5, 1, 2, 4, 8, 16, 64\}$, the best AG accuracy attained is 33.3% ($\alpha = 64$), still below Confidence-Select's 35.7%.

These negative results identify a structural distinction between *aggregation* (robust statistical combination across discriminators) and *selection* (delegation to a single discriminator). The unified framework is the multiplicative-VCG DSIC umbrella; the two regimes are distinguished by which mechanism within that umbrella is appropriate. Section 5 formalises this dichotomy.

Concrete evidence for the AG ceiling. Partition the 126 AG traces by the three discriminators' $\arg\max_t \hat{\theta}_t^k$: 18 traces are *unanimous* (all three agree), 64 are *2-of-3* (two agree), 44 are *all-differ* (all three distinct). On the 44 all-differ traces, no within-trace aggregation or selection mechanism we tested exceeds 27.3%, while Gemini alone achieves 36.4%. The 9pp gap is an information-theoretic ceiling on single-trace mechanisms whenever one discriminator is genuinely better and the other two disagree with it confidently.

1 Setup and Notation

We follow the companion theory note. A trace of length T consists of T sequential actions by agents from a set of M agents; the trace outcome is $W = 0$ (failure). For each step $t \in [T]$ there is an unobservable fix necessity $\theta_t \in [\varepsilon, 1-\varepsilon]$, and the binary ground truth $\boldsymbol{y} \in \{0,1\}^T$ marks a single mistake step $t^* \in [T]$. We have $K = 3$ heterogeneous LLM discriminators producing reports $\hat{\theta}_t^k \in [\varepsilon, 1-\varepsilon]$, read as $\hat{\theta}_t^k = P_k(\text{"Yes"})$ from a per-step yes/no prompt.

Benchmark. Who&When (Zhang et al., 2025):

- Hand-Crafted (HC): 58 traces, 55 with in-range ground-truth step, $T \in [5, 130]$ (median $T = 32$).
- Algorithm-Generated (AG): 126 traces, $T = 10$ for almost all.

Discriminators: D_0 = Gemini-3-flash, D_1 = Qwen3-Next-80B-A3B-Instruct, D_2 = GPT-OSS-120B; all at temperature 0.2 with the same prompt template.

Step accuracy. The primary metric is whether the mechanism's predicted mistake step $\hat{t}^* = \arg\max_t \bar{\theta}_t$ (where $\bar{\theta}_t$ is the mechanism's final per-step consensus) matches the ground truth t^*.

2 Empirical Diagnosis of the Original Mechanism

The original mechanism uses the value function

$$v_k^{\text{orig}}(\boldsymbol{d}, \hat{\theta}^k) = \prod_{t=1}^{T} \left[1 - (d_t - \hat{\theta}_t^k)^2\right].$$

With $R = 0$ and $\varepsilon = 10^{-6}$: $\text{acc}_{\text{HC}} = 9.1\%$, $\text{acc}_{\text{AG}} = 27.8\%$. Both are below all simple-aggregator baselines (Mean, Median, single discriminator). We identify two distinct causes.

Failure mode 1 (HC): heavy-tail confidently-wrong errors. On HC, D_0 assigns $\hat{\theta}_{t^*}^{D_0} < 0.01$ on 21.8% of traces: the model confidently flags the wrong step on roughly one trace in five. Under squared loss $(d_t - \hat{\theta}_t^k)^2$, a single such reading drives the multiplicative kernel factor close to zero and collapses D_k's cross-step reputation $w_k^t = \prod_{t' \neq t}[\,\cdot\,]$, in direct conflict with the intent of reputation-weighted aggregation.

Failure mode 2 (AG): one dominant discriminator. On AG, the average P(mistake step) per discriminator is $(0.64, 0.51, 0.65)$ but the rates at which the discriminator's $\arg\max_t$ coincides with t^* are $(38.1\%, 27.0\%, 19.0\%)$: D_2 gives *high* probabilities on the mistake step *and* on many others, while D_0's high probabilities are concentrated on the right step. Cross-step reputation conflates "frequent high probability" with "informative high probability", and the original mechanism penalises D_0's correct-but-isolated disagreements relative to the other two.

3 Proposed Mechanisms

We propose two mechanisms, each addressing one of the failure modes above, both within the multiplicative-VCG DSIC framework.

3.1 L1-Huber Multiplicative VCG (heavy-tail regime)

Definition 1 (L1-Huber kernel and value function)**.** *For fixed truncation $c \in (0, 1)$,*

$$\kappa_c(x) := 1 - \min(|x|, c), \qquad v_k^{L1H}(\boldsymbol{d}, \hat{\theta}^k) := \prod_{t=1}^{T} \kappa_c(d_t - \hat{\theta}_t^k).$$

Properties: $\kappa_c(0) = 1$, $\kappa_c(x) \in [1-c, 1]$, piecewise linear with one kink at $|x| = c$.

Reputation weight. Structurally unchanged:

$$w_k^t(\boldsymbol{d}) := \prod_{t' \neq t} \kappa_c(d_{t'} - \hat{\theta}_{t'}^k) = v_k^{\mathrm{L1H}}(\boldsymbol{d}, \hat{\theta}^k) / \kappa_c(d_t - \hat{\theta}_t^k).$$

Since $\kappa_c \geq 1 - c > 0$, the reputation never collapses to zero and all discriminators retain a non-trivial voice at every step.

Coordinate FOC: reputation-weighted median. Define the active set $A_t(\boldsymbol{d}) := \{k \in [K] : |d_t - \hat{\theta}_t^k| < c\}$. The directional derivative is

$$\partial_{d_t} V^{\mathrm{L1H}}(\boldsymbol{d}) = -\sum_{k \in A_t(\boldsymbol{d})} w_k^t(\boldsymbol{d}) \cdot \mathrm{sgn}(d_t - \hat{\theta}_t^k),$$

yielding the FOC

$$\boxed{d_t = \mathrm{WeightedMedian}_{k \in A_t(\boldsymbol{d})}\big(\hat{\theta}_t^k \;;\; w_k^t(\boldsymbol{d})\big).} \tag{1}$$

The single mechanism-design substitution mean $\to$ median is the entire content of the L1 kernel: the coordinate update is robust to a single high-weight outlier (squared-loss is not).

Algorithm 1 L1-Huber Multiplicative VCG (Phase 1b, $R = 0$)

Require: Reports $\hat{\theta} \in [\varepsilon, 1-\varepsilon]^{K\times T}$; bandwidth c; tolerance τ; max sweeps L.
Initialise $d_t \leftarrow \mathrm{median}_k \hat{\theta}_t^k$ for all t.
$V_{\mathrm{old}} \leftarrow \sum_k \prod_t \kappa_c(d_t - \hat{\theta}_t^k)$.
for sweep $\ell = 1, \ldots, L$ (Gauss–Seidel) **do**
  for $t = 1, \ldots, T$ **do**
    $w_k^t \leftarrow \prod_{t' \neq t} \kappa_c(d_{t'} - \hat{\theta}_{t'}^k)$ for all k.
    $A_t \leftarrow \{k : |d_t - \hat{\theta}_t^k| < c\}$.
    if $A_t \neq \emptyset$ **then**
      $d_t \leftarrow \mathrm{clip}\big(\mathrm{WeightedMedian}_{k\in A_t}(\hat{\theta}_t^k; w_k^t);\ \varepsilon, 1-\varepsilon\big)$.
    end if
  end for
  $V_{\mathrm{new}} \leftarrow \sum_k \prod_t \kappa_c(d_t - \hat{\theta}_t^k)$; if $V_{\mathrm{new}} - V_{\mathrm{old}} < \tau$: break; else $V_{\mathrm{old}} \leftarrow V_{\mathrm{new}}$.
end for
return $\hat{t}^* \leftarrow \arg\max_t d_t$.

Bandwidth. Empirical optimum on HC is $c = 0.05$. We verified non-singularity by scanning $c \in \{0.03, 0.05, 0.08, 0.10, 0.20, 0.50, 1.00\}$ (Appendix A). A data-adaptive choice $c = \mathrm{clip}(1.4826 \cdot \mathrm{MAD}(\hat{\theta}); 0.05, 0.95)$ hits the floor 0.05 on 100% of HC traces and yields identical accuracy.

3.2 Confidence-Based VCG Selection (dominance regime)

When one discriminator is substantially more accurate, per-step aggregation systematically dilutes its signal. We propose delegation:

Definition 2 (Confidence-based VCG selection)**.**

$$k^* := \arg\max_{k\in[K]} \max_{t\in[T]} \hat{\theta}_t^k, \qquad \hat{t}^* := \arg\max_{t\in[T]} \hat{\theta}_t^{k^*}.$$

Rationale. A peak probability close to 1 indicates concentrated mass on a small set of steps and is therefore discriminative at $\arg\max_t$. Among several selection criteria we explored — VCG payment ranking, mean L1-Huber reputation weight, L1-disagreement with the L1-Huber consensus, distributional entropy, peak-minus-mean, peak-separation, and Gini coefficient — peak confidence produced the highest AG accuracy. We discuss why it generalises well in Section 5.

Algorithm 2 Confidence-Based VCG Selection

Require: Reports $\hat{\theta} \in [\varepsilon, 1-\varepsilon]^{K\times T}$.
$s_k \leftarrow \max_t \hat{\theta}_t^k$ for all $k \in [K]$.
$k^* \leftarrow \arg\max_k s_k$; **return** $\arg\max_t \hat{\theta}_t^{k^*}$.

4 Empirical Results

Comparison to the $R = 5, \varepsilon = 0.10$ baseline. The earlier numbers Tong reported on AG used $R = 5$ OMD iterations together with aggressive clipping at $\varepsilon = 0.10$, yielding 34.9% on AG and 9.1% on HC. We report the original mechanism at $R = 0, \varepsilon = 10^{-6}$ as the clean "intrinsic VCG" baseline, because at $\varepsilon = 0.10$ the clipping squashes 78% of the input dynamic range and the mechanism degenerates to approximately Mean(prob) with $\varepsilon = 0.10$: indeed Mean(prob) at $\varepsilon = 0.10$ itself scores 34.9% on AG, identical to the $R = 5, \varepsilon = 0.10$ VCG. The $R = 0$ baseline isolates what the mechanism is intrinsically doing.

4.1 Main results with confidence intervals

Table 2: Step-attribution accuracy on Who&When. Bootstrap 95% CI in brackets ($B = 5{,}000$); McNemar exact p-value against original VCG. Significance: $^{**}p < 0.05$, $^{*}p < 0.10$.

Method	Hand-Crafted ($n = 55$) Acc. [95% CI]	Hand-Crafted ($n = 55$) p vs. orig.	Algorithm-Generated ($n = 126$) Acc. [95% CI]	Algorithm-Generated ($n = 126$) p vs. orig.
VCG-original (R=0)	9.1% [1.8, 16.4]	–	27.8% [19.8, 35.7]	–
Practical baselines (no oracle access):				
Mean (prob), $\varepsilon = 10^{-6}$	12.7% [5.5, 21.8]	–	33.3% [25.4, 41.3]	–
Mean (prob), $\varepsilon = 0.10$	10.9% [3.6, 20.0]	–	34.9% [27.0, 43.7]	–
Median (prob)	20.0% [9.1, 30.9]	0.11	29.4% [21.4, 37.3]	–
L1-Huber VCG ($c = 0.05$)	**23.6**% [12.7, 34.6]	**0.022****	30.2% [22.2, 38.1]	–
Confidence-Select VCG	14.5% [5.5, 23.6]	–	**35.7**% [27.0, 43.7]	**0.099***
Oracle ceiling (requires ground-truth selection):				
Best single discriminator	16.4% [7.3, 27.3]	–	38.1% [30.2, 46.8]	–

4.2 Per-regime breakdown

We classify each trace by the agreement structure of the $K = 3$ discriminators' $\arg\max_t$: Unanim. (all three agree), 2-of-3, All-differ.

Table 3: Per-regime step-attribution accuracy. Regime sizes in parentheses. **Bold** within column (excluding the oracle row).

	HC ($n = 55$)			AG ($n = 126$)		
	Unan. (5)	2-of-3 (17)	Differ (33)	Unan. (18)	2-of-3 (64)	Differ (44)
VCG-original (R=0)	20.0%	11.8%	6.1%	66.7%	18.8%	25.0%
Mean (prob)	**40.0**%	23.5%	3.0%	**66.7**%	28.1%	27.3%
Median (prob)	**40.0**%	35.3%	9.1%	**66.7**%	26.6%	18.2%
L1-Huber VCG ($c = 0.05$)	**40.0**%	**35.3**%	**15.2**%	**66.7**%	26.6%	20.5%
Confidence-Select VCG	**40.0**%	29.4%	3.0%	**66.7**%	**32.8**%	27.3%
Oracle Best Single (HC: D_2; AG: D_0)	40.0%	17.6%	12.1%	66.7%	31.2%	36.4%

HC: L1-Huber wins in the hardest regime. 60% of HC traces fall in the All-differ regime. On that regime, L1-Huber achieves 15.2% vs. 9.1% for the best baseline (Median) and 3.0% for Mean. The cross-step reputation-weighted median recovers signal that per-step aggregators miss; this is the L1-Huber mechanism doing real work where simple aggregation fails.

AG: the All-differ regime is a within-trace ceiling. On the 44 all-differ AG traces, no within-trace mechanism exceeds 27.3% while D_0 alone achieves 36.4%. This 9pp gap motivates the detailed analysis in Section 5.

5 Unification Attempts and Their Failure

The empirical pattern — L1-Huber wins HC, Confidence-Select wins AG — naturally raises the question: is there a single parametric mechanism whose behaviour smoothly interpolates between the two? We attempted two unifications. Both fail, and both failures are informative.

5.1 Cross-trace VCG-payment prior

The first attempt augments the per-trace reputation with a global discriminator prior π_k estimated from L1-Huber VCG payments aggregated across all other traces (leave-one-out, to avoid in-sample contamination):

$$\pi_k^{(i)} \propto \sum_{j \neq i} \Pi_k^{(j)}, \qquad w_k^t \to \pi_k^{(i)} \cdot \prod_{t' \neq t} \kappa_c(d_{t'} - \hat{\theta}_{t'}^k).$$

We test both π-*select* (predict $\arg\max_t \hat{\theta}_t^{k^*}$ where $k^* = \arg\max_k \pi_k$) and π-*weighted L1-Huber* (multiplicative augmentation of the reputation).

Table 4: Cross-trace aggregated statistics on AG ($n = 126$). All three aggregation methods (sum of payments, mean payment, mean cross-step reputation) nominate D_1 (Qwen) as the highest-prior discriminator — but D_0 (Gemini) is the most *accurate* discriminator. The prior inverts truth.

	D_0 Gemini	D_1 Qwen	D_2 GPT-OSS
Sum of L1-Huber payments	4.89	**11.16**	3.10
Mean payment	0.039	**0.089**	0.025
Mean cross-step reputation	0.811	**0.881**	0.817
Fraction of traces $\Pi_k > 0$	0.762	**0.937**	0.770
Single argmax-step accuracy	**38.1**%	27.0%	19.0%

The result on AG: π-select yields 27.0% (exactly Qwen's solo accuracy — the prior consistently nominates Qwen). π-weighted L1-Huber yields 29–31%, indistinguishable from the L1-Huber baseline at 30.2%.

Proposition 3 (informal)**.** *The VCG payment $\Pi_k = S_{-k}(\boldsymbol{d}^*_{-k}) - S_{-k}(\boldsymbol{d}^*)$ measures how much removing k shifts the consensus relative to the others' welfare. A discriminator who reports values close to the cross-discriminator median on most steps has high welfare contribution to the others; a lone correct expert who reports values that disagree with the others' false consensus has lower welfare contribution. Consequently, any reputation prior built from a consensus-based payment is biased toward consensus-followers and away from minority correct experts.*

This impossibility is structural: payment-based priors cannot resolve the dominance regime regardless of the kernel used.

5.2 Within-trace evidence prior

The second attempt avoids cross-trace contamination entirely. The prior π_k depends only on the current trace's reports:

$$w_k^t = \pi_k(\hat{\theta})^\alpha \cdot \prod_{t' \neq t} \kappa_c(d_{t'} - \hat{\theta}_{t'}^k),$$

with π_k derived from intrinsic peakiness measures:

$$\pi_k^{\text{peak}} := \max_t \hat{\theta}_t^k, \quad \pi_k^{\text{peaksep}} := \hat{\theta}_{k,\cdot}^{(1)} - \hat{\theta}_{k,\cdot}^{(2)}, \quad \pi_k^{\text{negent}} := \exp(-H_k).$$

$\alpha = 0$ recovers L1-Huber; large α approaches hard selection on π.

Table 5: Bandwidth-free α scan of the peak-prior soft-augmented L1-Huber on both subsets. The mechanism cannot simultaneously match L1-Huber on HC and Confidence-Select on AG.

α	HC acc. (peak)	HC acc. (negent)	AG acc. (peak)	AG acc. (negent)
0	**23.6**%	**23.6**%	30.2%	30.2%
0.5	23.6%	20.0%	29.4%	30.2%
1	23.6%	20.0%	30.2%	28.6%
2	23.6%	20.0%	31.0%	27.0%
4	20.0%	18.2%	29.4%	27.8%
8	18.2%	16.4%	29.4%	27.8%
16	16.4%	16.4%	27.8%	27.8%
64	16.4%	14.5%	33.3%	27.8%

The pattern is clear: *no setting of* α *matches Confidence-Select's* 35.7% *on AG, and* $\alpha > 0$ *never strictly improves HC over L1-Huber.* The two best mechanisms are not parametric neighbours.

5.3 The aggregation–selection dichotomy

The two negative results identify a structural distinction.

Aggregation (L1-Huber Eq. 1): the coordinate update places d_t at the multi-discriminator reputation-weighted median, conditional on each k's self-reported probability at step t. The mass of any single discriminator's high report is diluted by the others.

Selection (Confidence-Select): the decision uses only the chosen discriminator's internal $\arg\max_t$. The other discriminators' votes are *ignored*, not weighted.

A soft prior cannot replicate the structural change from one mode to the other: as $\pi_{k^*} \to \infty$ in π-weighted L1-Huber, the weighted-median update converges to k^*'s report *at the current step* t, but $\arg\max_t d_t$ still depends on the full allocation trajectory, not on k^*'s argmax. Even in this limit, the final prediction differs from Confidence-Select's. The empirical observation that $\alpha = 64$ on AG yields 33.3% rather than 35.7% is consistent with this analysis.

Remark 4 (Practical regime selection)**.** *We therefore characterise the unified VCG framework as comprising* two regime-specific mechanisms *subsumed under one DSIC umbrella, rather than a single parametric mechanism. The appropriate mechanism is selected by a simple data-level rule:* long traces or per-trace report concentration is low $\Rightarrow$ *Aggregation (L1-Huber);* short traces or per-trace report concentration is high $\Rightarrow$ *Selection (Confidence-Select). This is analogous to robust regression, where Huber loss and Tukey biweight both belong to the M-estimator family but are non-interpolating, with the appropriate choice determined by the noise distribution.*

6 Theoretical Extension: Anchored Correctness for L1-Huber

The theory note's main result states that under (f, γ, η)-anchoring of the discriminator reports and DSIC, the original multiplicative VCG identifies the correct mistake step with probability tending to one. We sketch the analogue for L1-Huber with strictly cleaner constants.

Theorem 5 (L1-Huber Anchored Correctness, sketch)**.** *Suppose the discriminator reports are* (f, γ, η)*-anchored with* $f \leq \lfloor (K-1)/2 \rfloor$ *and* $\eta < c$*, and the margin conditions* $m_{\text{thr}}, m_{\text{step}}, m_{\text{agent}} > 0$ *of the theory note hold. Let* $\boldsymbol{d}^*$ *be the L1-Huber allocation (Algorithm 1). Then there exist constants* $C_1, c' > 0$ *depending only on* (K, ε, c) *such that for every* $t \in [T]$*,*

$$|d_t^* - \theta_t| \;\leq\; \eta + C_1 \exp\big(- c' \, \gamma \, (T-1) \big),$$

and the attribution decisions $(S^*, \hat{t}^*, \hat{\imath}^*)$ *are correct whenever the right-hand side is less than the corresponding margin.*

Proof idea. *Step 1* (anchor weights bounded below): on $\gamma(T-1)$ anchored steps, anchor factor $\geq 1 - \eta - c$; on remaining steps, factor $\geq 1 - c$. Combined: $w_k^t \geq (1-c)^{(1-\gamma)(T-1)}(1-\eta-c)^{\gamma(T-1)}$, depending only on (c, η, γ) — not on report scale. *Step 2* (non-anchor weights decay exponentially): non-anchor disagreement at rate $\geq \gamma_{\mathcal{B}} > 0$ produces $w_k^t \leq (1-c)^{\gamma_{\mathcal{B}}(T-1)}$. *Step 3* (median robustness): the weighted-median FOC is robust to any coalition of combined weight $< 1/2$, strictly stronger than the weighted mean which is biased by an arbitrary amount via one high-weight outlier.

Remark 6 (Where L1-Huber is strictly stronger than squared loss)**.** *The original mechanism's anchor lower bound depends on the clipped report deviation magnitudes (through* $1 - (1-2\varepsilon)^2$*). The L1-Huber bound depends only on* (c, η, γ)*, decoupling robustness from the report scale. This is the formal counterpart of the empirical observation that L1-Huber outperforms the original and Huber (squared truncated) variants on heavy-tail HC.*

Remark 7 (Where Theorem 5 does not apply)**.** *The theorem requires an anchor set of size* $K - f \geq \lceil K/2 \rceil$*. Under* $K = 3$ *with* $f = 2$*, the anchor set is a single discriminator and the hypothesis fails. This is precisely the AG situation: one discriminator dominates, the other two are non-anchors, and no within-trace mechanism can identify the dominant one from cross-step consistency alone — as demonstrated empirically in Section 5.*

7 Limitations and Open Questions

What we have not proved. Section 5's Proposition 3 is informal. A formal statement would specify: for any monotone increasing Π in cross-step agreement with consensus, there exists a report distribution under which the Π-induced prior selects the wrong discriminator. We believe this is provable but defer it to a follow-up.

Larger K**.** With $K = 3$ and $f = 2$, AG falls outside the anchored regime by one discriminator. With $K = 5$ or $K = 7$ the anchoring hypothesis becomes much more plausible, and the dominance regime may itself become an anchored regime under a larger discriminator pool. Re-running the AG experiments with two additional discriminators is the most direct next test.

OMD with the L1-Huber kernel. The OMD calibration of Phase 2 was not integrated with L1-Huber because the finite-difference gradient is kernel-dependent. A kernel-aware implementation could provide additional gain over the $R = 0$ results.

Other selection criteria. We use peak confidence in Confidence-Select; the criterion is empirical, not derived from a formal IC argument. A more principled selection rule (e.g. posterior probability of being the unique anchor in a generative model of the reports) would strengthen the AG side of the story.

8 Discussion Points for Collaborator Review

- Should the AG experiments be re-run with $K = 5$ before submission?
- Theorem 5 full proof: main paper, appendix, or companion theory-note revision?
- Should we attempt a formal statement of Proposition 3 for inclusion?
- Bandwidth c: single value or MAD-based selector as the canonical choice?
- OMD with L1-Huber: include in this submission or defer?

A Bandwidth Sensitivity for L1-Huber on HC

c	HC step accuracy
0.03	20.0% (11/55)
0.05	**23.6**% (13/55)
0.08	18.2% (10/55)
0.10	16.4% (9/55)
0.20	14.5% (8/55)
0.50	18.2% (10/55)
1.00	12.7% (7/55)

Data-adaptive $c = \text{clip}(1.4826 \cdot \text{MAD}(\hat{\theta}); 0.05, 0.95)$ hits the floor 0.05 on 100% of HC traces.

B Reproducibility

All numbers reproduce with the scripts in the project repository:

- Main accuracy and per-regime breakdown: `scripts/explore_ag.py`
- Bootstrap CIs and McNemar tests: `scripts/bootstrap_ci.py`
- Bandwidth scan: `scripts/run_ablation_huber.py`
- Cross-trace prior experiment: `scripts/cross_trace_prior.py`
- Within-trace prior scan: `scripts/unified_mechanism.py`

L1-Huber kernel in `vcg/allocation_l1huber.py`; random seed 42 throughout; $B = 5{,}000$ bootstrap resamples in all CIs and p-values.